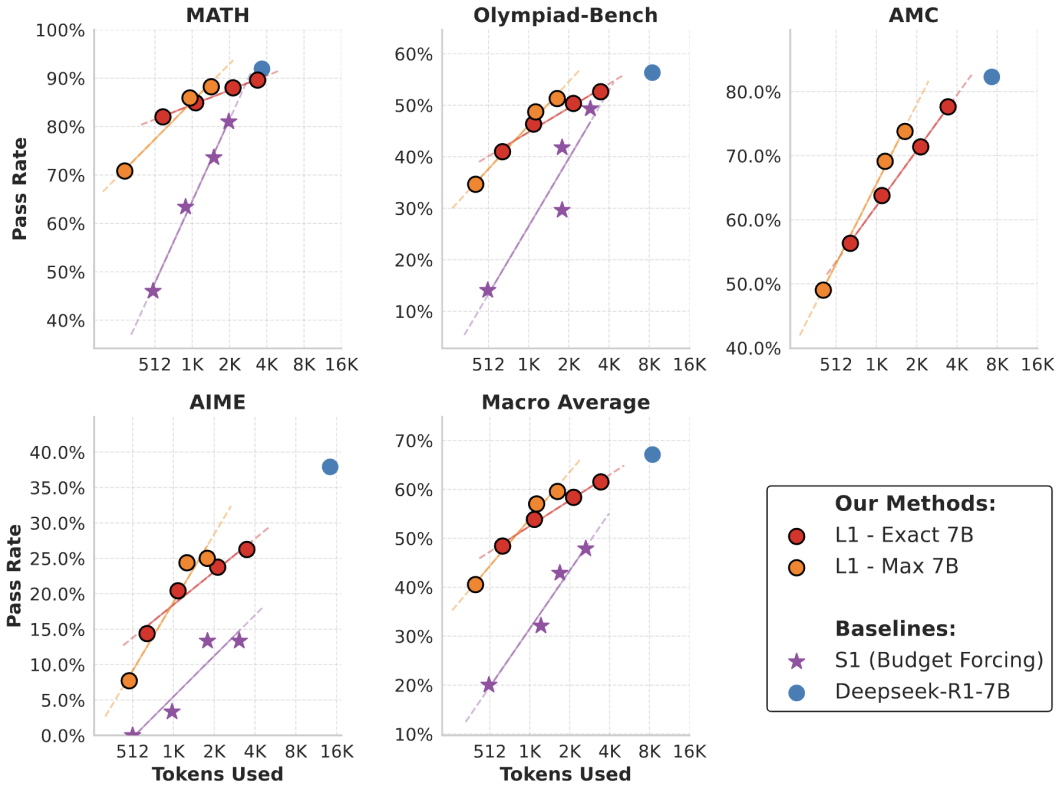

Figure 19: **Scaling LCPO to a 7B reasoning model.** We apply LCPO to DeepSeek-R1-Distill-7B reasoning model and show strong length control and performance on math reasoning tasks.

Example

Question

Find the largest possible real part of $[(75+117i)z + \frac{96+144i}{z}]$ where z is a complex number with $|z| = 4$. Let's think step by step and output the final answer within boxed{}. Think for 512 tokens.

Correct Answer

$\boxed{540}$

Correctness

✗ **Incorrect**

Tokens Requested

512

Model Response

Thinking Process

Okay, so I have this problem: I need to find the largest possible real part of the expression $(75+117i)z + (96+144i)/z$, where z is a complex number with $|z| = 4$. Hmm, let's break this down.
First, since $|z| = 4$, z can be written as $4e^{i\theta}$ for some angle θ. That might help simplify things.
Let me denote $z = 4e^{i\theta}$. Then, $1/z = (1/4)e^{-i\theta}$ because $|z| = 4$.
So, the expression becomes $(75+117i) \cdot 4e^{i\theta} + (96+144i) \cdot (1/4)e^{-i\theta}$.
Simplifying, that's $4(75+117i)e^{i\theta} + (24+36i)e^{-i\theta}$.
Now, I need to find the real part of this expression. Let's denote $\text{Re}[(75+117i)z + (96+144i)/z]$.
Breaking it down, $\text{Re}[(75+117i)z] + \text{Re}[(96+144i)/z]$.
Since $z = 4e^{i\theta}$, $\text{Re}[(75+117i) \cdot 4e^{i\theta}] + \text{Re}[(96+144i) \cdot (1/4)e^{-i\theta}$.
This becomes $4\text{Re}[(75+117i)e^{i\theta}] + (1/4)\text{Re}[(96+144i)e^{-i\theta}]$.
Now, let's compute $\text{Re}[(75+117i)e^{i\theta}]$. This is $75\cos\theta + 117\sin\theta$.
Similarly, $\text{Re}[(96+144i)e^{-i\theta}]$ is $96\cos\theta - 144\sin\theta$.
So, putting it all together, the real part is $4(75\cos\theta + 117\sin\theta) + (1/4)(96\cos\theta - 144\sin\theta)$.
Simplifying, that's $300\cos\theta + 468\sin\theta + 24\cos\theta - 36\sin\theta$.
Combining like terms, we get $(300+24)\cos\theta + (468-36)\sin\theta = 324\cos\theta + 432\sin\theta$.
Now, we need to find the maximum of $324\cos\theta + 432\sin\theta$. This can be written as $R\cos(\theta - \phi)$, where $R = \sqrt{324^2 + 432^2}$.
Calculating R: $324^2 = 104,976$, $432^2 = 186,624$. Sum is $391,600$. $R = \sqrt{391,600} \approx 625.6$.
Thus, the maximum real part is approximately 625.6.

The largest possible real part is $\boxed{625.6}$.

Figure 20: Example of model response with 512 tokens. Note that the model's answer is incorrect.

Example

Question

Find the largest possible real part of $[(75+117i)z + \frac{96+144i}{z}]$ where z is a complex number with $|z| = 4$. Let's think step by step and output the final answer within $boxed\{\}$. Think for 3600 tokens.

Correct Answer

$\boxed{540}$

Correctness

✔ **Correct**

Tokens Requested

3600

Model Response

Thinking Process (not shown to user)

Okay, so I have this problem: I need to find the largest possible real part of the expression (75 + 117i)z + (96 + 144i)/z, where z is a complex number with $|z| = 4$. Hmm, that seems a bit involved, but let's break it down step by step. First, let me understand what is given. We have a complex number z with magnitude 4, so z can be written in the form z = $4e^{i\theta}$, where θ is the argument of z. That might be useful because it allows us to express z in terms of its angle, which could simplify the expression.
The expression we need to consider is $(75+117i)z + \frac{96+144i}{z}$. We are asked to find the largest possible real part of this expression. So, let's denote the expression as $E = (75+117i)z + \frac{96+144i}{z}$.
[Reasoning process truncated for brevity ...]
Then, adding $104,976 + 186,624 = 291,600$. $\sqrt{291,600} = \sqrt{2916 * 100} = \sqrt{2916} * 10$. Now, $\sqrt{2916}$: let's see, $54^2 = 2916$, because $50^2 = 2500$, $54^2 = 2916$. So, $\sqrt{2916} = 54$, thus $\sqrt{2916 * 100} = 54 * 10 = 540$. So, that seems correct.
Therefore, the maximum value of $\mathrm{Re}(E)$ is 540.
Therefore, the largest possible real part is 540.

Given the expression $(75+117i)z + \frac{96+144i}{z}$ where z is a complex number with $|z| = 4$, we need to find the largest possible real part of this expression.
First, we express z as $4e^{i\theta}$ since $|z| = 4$. Then, we substitute z and $\frac{1}{z}$ into the expression:

$$(75+117i)z + \frac{96+144i}{z}$$

[Output truncated for brevity ...]
Thus, the largest possible real part is:

$$\boxed{540}$$

Figure 21: Example of model response with 3600 tokens. With more tokens, the model correctly calculates the answer.